

# Recursive trees grown from dynamic building sequences

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## Abstract

We study the depth  $D_n$  of node  $n$  in a preferential recursive tree grown under dynamic affinities with age index  $\Lambda \in [0, 2]$ . Exact and asymptotic formulae for the mean and the variance are derived. As a consequence,  $D_n/\ln n$  converges in probability to  $2/(1 + \Lambda)$ . We then establish the central limit theorem

$$\frac{D_n - \mathbb{E}[D_n]}{\sqrt{\ln n}} \xrightarrow{\mathcal{L}} \mathcal{N}(0, V_\Lambda), \quad V_\Lambda = \frac{2 + 8\Lambda - 2\Lambda^2}{(\Lambda + 1)^3}.$$

The proof of asymptotic normality proceeds from the parent-index decomposition of the centred depth, an averaging lemma for the attachment kernel, and convergence of all moments via a mixture recurrence.

**Keywords:** random recursive tree; preferential attachment; dynamic affinities; depth; central limit theorem.

## 1 Introduction

The recursive tree is a classical random structure. Early applications appear in the analysis of pyramid schemes [7] and in philology [13].

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Starting with one node at time 0, at each new tick of discrete time, a new node appears and is adjoined by an edge to an existing tree node, which we call the *parent*. Each node is labeled with its order of appearance in the tree. The operation of letting a parent acquire a new node (*child*) is often viewed as *recruiting*.

So, the initial node is the first in the structure and takes the label 1, the second node to appear takes the label 2, and so on. Generally, the  $n$ th appearance gives the node generated at that point the label  $n$ . Note the unit shift between the time of appearance and the label given to the node born at this time.

When the node labeled  $n$  joins the tree, there are  $n - 1$  nodes, any of which may become the parent. So, we consider a probability distribution on the set  $\{1, \dots, n - 1\}$  to parent the node labeled  $n$ , and each such distribution corresponds to an insertion algorithm. No matter what insertion algorithm (emerging from a specific probability model) is used, it induces a property along the labels on root-to-leaf paths: The labels on such a path are in an increasing order. Therefore, several authors call these structures “increasing trees” (see [3], for example).

Nakata and Mahmoud [12] considers a rather general probability model, in which a static arbitrary sequence  $\{a_k\}_{k=1}^{\infty}$  of positive real numbers is determined. They call  $\{a_k\}_{k=1}^{\infty}$  the *affinity sequence*, and consider  $a_k$  the affinity of node  $k$  after its appearance in the tree. For  $n \geq 2$ , let  $\mathcal{A}_{n,k}$  be the event that node  $k$  is the chosen parent for node  $n$  (when the tree size is  $n - 1$ ). The affinities are the driving force behind the probability model; affinities are turned into probabilities by dividing by  $s_{n-1} := \sum_{k=1}^{n-1} a_k$ . In other words, the probability model is

$$\mathbb{P}(\mathcal{A}_{n,k}) = \frac{a_k}{s_{n-1}}, \quad \text{for } k = 1, \dots, n - 1.$$

This probability model represents an entire class of recursive tree insertion algorithms with general affinities. It is a static preferential model with  $a_k$  specified ahead of time and remains the same throughout the life of the tree after time  $k - 1$ ; a node recruits according to a probability proportional to its affinity.

While the model in [12] is broad and captures many known special cases, the accelerating innovation in technology is dictating the need for additional growth models. The model in [12] covers the standard (uniform) recursive tree [4, 6, 9, 15], the young-age recruiting tree [10] and the power-weight

tree [11]. However, such a static affinity model does not cover a case in which the affinity of node  $k$  is a function of both  $k$  and  $n$ . There is need to consider affinities of the form  $a_{n,k}$ . For instance, the old-age model with affinities  $a_{n,k} = n - k$  stands for a rather natural application, where the nodes with smaller indices (older in the tree) have higher recruiting probabilities. Such a dynamic affinity takes “experience” into account.

To be specific, in this manuscript we study the recursive tree with the dynamic affinities

$$a_{n,k} = \alpha n + \beta k > 0, \quad \text{for } k = 1, \dots, n-1, \quad (1)$$

subject to  $2\alpha + \beta > 0$ . Writing  $s_{n-1} = \sum_{k=1}^{n-1} a_{n,k}$ , the attachment probabilities are

$$\mathbb{P}(\mathcal{A}_{n,k}) = \frac{a_{n,k}}{s_{n-1}}, \quad \text{for } k = 1, \dots, n-1.$$

The hypothesis  $2\alpha + \beta > 0$  ensures that  $s_{n-1}$  is positive and strictly increasing in  $n$ .

It is convenient to reduce the two parameters to a single *age index*

$$\Lambda = \frac{2\alpha}{2\alpha + \beta},$$

which ranges over the interval  $[0, 2]$ . In these coordinates, the event  $\mathcal{A}_{n,k}$  (node  $k$  recruits node  $n$ ) has probability

$$\mathbb{P}(\mathcal{A}_{n,k}) = \frac{a_{n,k}}{s_{n-1}} = \frac{\alpha n + \beta k}{(2\alpha + \beta)n(n-1)/2} = \frac{\Lambda n + 2(1 - \Lambda)k}{n(n-1)},$$

for  $k = 1, \dots, n-1$ .

**Remark 1.1.** *The dynamic affinities in (1) contain the following models:*

- **(The young-age model)** Take  $\alpha = 0$  and  $\beta > 0$  (hence  $\Lambda = 0$ ). The affinities are  $a_{n,k} = \beta k$ , and we get the probabilities

$$\mathbb{P}(\mathcal{A}_{n,k}) = \frac{2k}{n(n-1)}, \quad \text{for } k = 1, \dots, n-1.$$

*This model appears in [10, 11]. In this model, a node with a high index  $k$  (in a tree of size  $n$ ), has not lived long in the tree. It is still young, as opposed to nodes that appeared early, such as the root, which is now the oldest in the tree.*

- **(The uniform model)** Take  $\alpha > 0$  and  $\beta = 0$  (hence  $\Lambda = 1$ ). The affinities are  $a_{n,k} = \alpha n$ , to get the uniform probabilities

$$\mathbb{P}(\mathcal{A}_{n,k}) = \frac{1}{n-1}, \quad \text{for } k = 1, \dots, n-1.$$

In this model, all nodes in a tree of size  $n$  have the same affinity (same attractiveness). The model has been studied extensively; see [4, 6, 15].

- **(The old-age model)** Take  $\alpha > 0$  and  $\beta = -\alpha$  (hence  $\Lambda = 2$ ). The affinities are  $a_{n,k} = \alpha(n-k)$ , and we get the probabilities

$$\mathbb{P}(\mathcal{A}_{n,k}) = \frac{2(n-k)}{n(n-1)}, \quad \text{for } k = 1, \dots, n-1.$$

This model appears in [9]. In this model, an early node with a low index  $k$  (in a tree of size  $n$ ), has lived long in the tree. It is now old, with high affinity.

Figure 1 depicts the six recursive old-age preferential trees of size 4. The numbers above the trees are their probabilities.

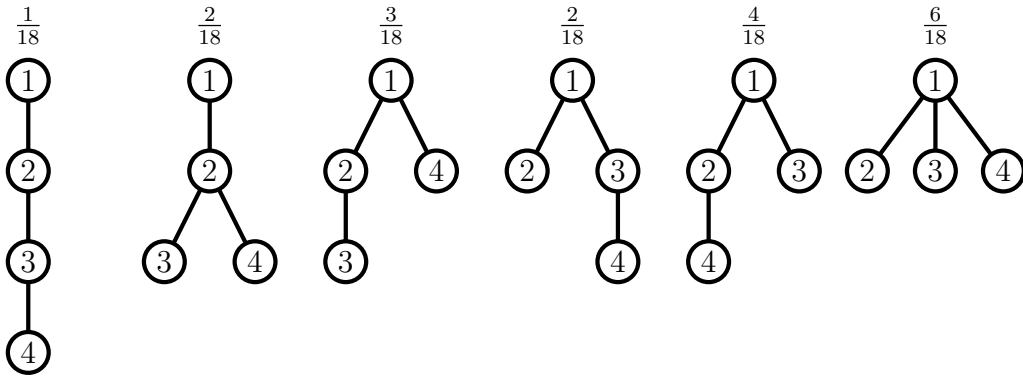


Figure 1: The old-age preferential recursive trees of size 4 and their probabilities. The affinity of node  $k$  is  $2(n-k)/(n(n-1))$ .

## 2 Notation

The indicator of an event  $\mathcal{E}$  is denoted by  $\mathbb{I}_{\mathcal{E}}$ . We use Iverson's brackets: for a predicate  $\mathcal{B}$ , the symbol  $[[\mathcal{B}]]$  equals 1 if  $\mathcal{B}$  is true and 0 otherwise. This notation is used to suppress terms that are absent for particular values of the parameters. For any expression  $g$ , we adopt the convention that  $[[\mathcal{B}]]g = 0$  whenever  $\mathcal{B}$  is false, even if  $g$  is formally undefined or infinite.

Several results are expressed in terms of harmonic numbers. For an integer  $r \geq 1$ , the sum  $\sum_{k=1}^n 1/k^r := H_n^{(r)}$  is the  $n$ th harmonic number of order  $r$ . It is customary to drop the superscript for the first-order harmonic number and to write  $H_n$  in place of  $H_n^{(1)}$ . The harmonic numbers of the first two orders appear in the exact mean and the exact variance; they admit the classical expansion

$$H_n = \ln n + \gamma + O\left(\frac{1}{n}\right), \quad (2)$$

where  $\gamma = 0.57721\dots$  is the *Euler–Mascheroni constant*.

The symbol  $\xrightarrow{\mathbb{P}}$  denotes convergence in probability, and  $\xrightarrow{\mathcal{L}}$  denotes convergence in distribution.

## 3 Depth

Let  $D_n$  be the depth of node  $n$  in a recursive tree grown under preferential dynamic affinities. In this section we derive exact and asymptotic formulae for the mean and the variance of  $D_n$ . The resulting growth rates yield a weak law of large numbers by Chebyshev's inequality. The limiting distribution is established in Subsection 3.3.

Let  $\psi_n(t) = \mathbb{E}[e^{D_n t}]$  be the moment generating function of  $D_n$ . For  $n \geq 2$ , the depth satisfies the recursion

$$D_n = \sum_{k=1}^{n-1} (1 + D_k) \mathbb{I}_{\mathcal{A}_{n,k}}. \quad (3)$$

In the present model the affinities depend only on the labels, so  $\mathcal{A}_{n,k}$  is independent of  $D_k$ . Consequently, the sequence  $\psi_n(t)$  obeys

$$\psi_n(t) = \sum_{k=1}^{n-1} \mathbb{E}[e^{(1+D_k)t}] \mathbb{P}(\mathcal{A}_{n,k})$$

$$\begin{aligned}
&= e^t \sum_{k=1}^{n-1} \mathbb{E} [e^{D_k t}] \mathbb{P}(\mathcal{A}_{n,k}) \\
&= \frac{e^t}{n(n-1)} \sum_{k=1}^{n-1} (\Lambda n + 2(1-\Lambda)k) \psi_k(t). \tag{4}
\end{aligned}$$

It is convenient to write  $n(n-1) = 2\binom{n}{2}$ ; we adopt this abbreviation throughout.

The recurrence (4) requires two successive differencing steps in order to eliminate the cumulative sums. This is the principal technical difference with the static-affinity models treated in [12], which typically require only one differencing step.

#### **The first step of differencing:**

From (4), for  $n \geq 2$  we have

$$\binom{n}{2} \psi_n(t) = e^t \sum_{k=1}^{n-1} \left( \frac{\Lambda}{2} n + (1-\Lambda)k \right) \psi_k(t).$$

Differencing requires that the formula be true for both  $n$  and  $n-1$ . So, the least  $n$  to consider is 3. By differencing, for  $n \geq 3$  we obtain

$$\begin{aligned}
&\binom{n}{2} \psi_n(t) - \binom{n-1}{2} \psi_{n-1}(t) \\
&= e^t \left( \left( \left( 1 - \frac{\Lambda}{2} \right) n - (1-\Lambda) \right) \psi_{n-1}(t) + \frac{\Lambda}{2} \sum_{k=1}^{n-2} \psi_k(t) \right).
\end{aligned}$$

We observe that the formula holds for  $n \geq 2$ , as well.

#### **The second step of differencing:**

Difference again. After simplification, we obtain

$$\begin{aligned}
&\binom{n}{2} \psi_n(t) - \left( 2\binom{n-1}{2} + e^t \left( \left( 1 - \frac{\Lambda}{2} \right) n - (1-\Lambda) \right) \right) \psi_{n-1}(t) \\
&+ \left( \binom{n-2}{2} + e^t \left( 1 - \frac{\Lambda}{2} \right) (n-2) \right) \psi_{n-2}(t) = 0, \quad \text{for } n \geq 3. \tag{5}
\end{aligned}$$

### 3.1 The mean depth

Taking derivatives of (5) with respect to  $t$  and evaluating at  $t = 0$ , for all  $n \geq 2$ , we get the recurrence:

$$\binom{n}{2} \mathbb{E}[D_n] - \left\{ 2 \binom{n-1}{2} + \left(1 - \frac{\Lambda}{2}\right) n - (1 - \Lambda) \right\} \mathbb{E}[D_{n-1}] + \left\{ \binom{n-2}{2} + \left(1 - \frac{\Lambda}{2}\right) (n-2) \right\} \mathbb{E}[D_{n-2}] = 1. \quad (6)$$

In the solution to this recurrence, certain sums of ratios of gamma functions appear. Starting from

$$\frac{\Gamma(i)}{\Gamma(i-\Lambda)} = \begin{cases} (i-1)(i-2), & \text{if } \Lambda = 2; \\ \frac{1}{\Lambda+1} \left( \frac{\Gamma(i+1)}{\Gamma(i-\Lambda)} - \frac{\Gamma(i)}{\Gamma(i-\Lambda-1)} \right), & \text{if } \Lambda \neq 2, \end{cases}$$

the sum on the index  $i = 3, \dots, n$  is telescopic, yielding

$$\sum_{i=3}^n \frac{\Gamma(i)}{\Gamma(i-\Lambda)} = \frac{\Gamma(n+1)}{(\Lambda+1)\Gamma(n-\Lambda)} - [[\Lambda \neq 2]] \frac{2}{(\Lambda+1)\Gamma(2-\Lambda)}. \quad (7)$$

**Remark 3.1.** *Many identities similar to (7) with sums of ratios of gamma functions are known. We are not certain what the best source for this identity is, so we refer to the appendix in the recent work [16], where the author proves several identities of this type.*

**Proposition 3.1.** *Let  $D_n$  be the depth of node  $n$  in a preferential recursive tree of age index  $\Lambda$ . For  $n \geq 2$ , we have*

$$\mathbb{E}[D_n] = \frac{2}{\Lambda+1} H_{n-1} + \frac{\Lambda-1}{\Lambda+1} + [[\Lambda \neq 2]] \frac{2(\Lambda-1)}{(\Lambda+1)\Gamma(2-\Lambda)} \sum_{i=3}^n \frac{\Gamma(i-\Lambda)}{(i-1)\Gamma(i+1)}.$$

*Proof.* Since  $D_1 = 0$ , we assume  $n \geq 2$ . In order to obtain an exact expression for  $\mathbb{E}[D_n]$ , we use the identity

$$2 \binom{n-1}{2} + \left(1 - \frac{\Lambda}{2}\right) n - (1 - \Lambda) = \binom{n}{2} + \left( \binom{n-2}{2} + \left(1 - \frac{\Lambda}{2}\right) (n-2) \right), \quad (8)$$

(valid for  $n \geq 2$ ) to create one extra layer of differences (valid for  $n \geq 3$ ). We write (6) in the form

$$\binom{n}{2} (\mathbb{E}[D_n] - \mathbb{E}[D_{n-1}])$$

$$- \left\{ \binom{n-2}{2} + \left(1 - \frac{\Lambda}{2}\right)(n-2) \right\} (\mathbb{E}[D_{n-1}] - \mathbb{E}[D_{n-2}]) = 1.$$

Now, put  $a_n = \mathbb{E}[D_n] - \mathbb{E}[D_{n-1}]$ , to see that for  $n \geq 3$ , we have the recurrence

$$a_n = \frac{(n-2)(n-1-\Lambda)}{n(n-1)} a_{n-1} + \frac{2}{n(n-1)}.$$

We note that the recurrence extends down to  $n = 2$ . We then solve the recurrence with the boundary condition  $a_2 = 1$ . This is a standard linear recurrence in the form

$$a_n = g_n a_{n-1} + h_n, \quad \text{for } n \geq 2, \quad (9)$$

where

$$g_n = \frac{(n-2)(n-1-\Lambda)}{n(n-1)} \quad \text{and} \quad h_n = \frac{2}{n(n-1)}.$$

Since<sup>5</sup>

$$\begin{aligned} a_n &= \sum_{i=2}^n h_i \prod_{j=i+1}^n g_j + a_1 \prod_{j=3}^n g_j \\ &= \sum_{i=2}^n \frac{2}{i(i-1)} \prod_{j=i+1}^n \frac{(j-2)(j-1-\Lambda)}{j(j-1)} \\ &= \prod_{j=3}^n \frac{(j-2)(j-1-\Lambda)}{j(j-1)} + \frac{2}{n-1} \sum_{i=3}^n \frac{(i-\Lambda)(i+1-\Lambda) \cdots (n-1-\Lambda)}{i(i+1) \cdots n}, \end{aligned}$$

We now plug in the identity (7):

$$\begin{aligned} a_n &= [[\Lambda \neq 2]] \frac{2\Gamma(n-\Lambda)}{\Gamma(2-\Lambda)(n-1)\Gamma(n+1)} \\ &\quad + \frac{2\Gamma(n-\Lambda)}{(n-1)\Gamma(n+1)} \left( \frac{\Gamma(n+1)}{(\Lambda+1)\Gamma(n-\Lambda)} - [[\Lambda \neq 2]] \frac{2}{(\Lambda+1)\Gamma(2-\Lambda)} \right) \\ &= [[\Lambda \neq 2]] \frac{2\Gamma(n-\Lambda)}{\Gamma(2-\Lambda)(n-1)\Gamma(n+1)} \\ &\quad + \frac{2}{(\Lambda+1)(n-1)} - [[\Lambda \neq 2]] \frac{4\Gamma(n-\Lambda)}{(\Lambda+1)\Gamma(2-\Lambda)(n-1)\Gamma(n+1)}, \end{aligned}$$

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<sup>5</sup>As customary, interpret the product as 1, when the lower index is  $n+1$ .

valid for  $n \geq 3$ .

$$a_n = \frac{2}{(n-1)(\Lambda+1)} + [[\Lambda \neq 2]] \frac{2(\Lambda-1)\Gamma(n-\Lambda)}{(\Lambda+1)\Gamma(2-\Lambda)(n-1)\Gamma(n+1)}.$$

Recall that  $a_n = \mathbb{E}[D_n] - \mathbb{E}[D_{n-1}]$  and  $\mathbb{E}[D_2] = 1$ , and unwind the recurrence to yield

$$\mathbb{E}[D_n] = 1 + \frac{2}{\Lambda+1}(H_{n-1}-1) + [[\Lambda \neq 2]] \frac{2(\Lambda-1)}{(\Lambda+1)\Gamma(2-\Lambda)} \sum_{i=3}^n \frac{\Gamma(i-\Lambda)}{(i-1)\Gamma(i+1)}.$$

□

**Remark 3.2.** *The expression for the mean in Proposition 3.1 is reduced to a closed form if  $\Lambda$  is an integer. Simply in this case the summand in the remaining sum is a product of integers. With  $\Lambda \in [0, 2]$ , the only relevant integers are 0, 1, and 2, corresponding to young-age preferential trees, uniform recursive trees and old-age preferential trees. Corollaries 3.1–3.3 illustrate this point.*

**Corollary 3.1.** *In a young-age preferential recursive tree, for  $n \geq 3$ , we have*

$$\mathbb{E}[D_n] = 2(H_n - 1).$$

*Proof.* In a young-age recursive preferential tree, we have  $\alpha = 0$ ,  $\beta > 0$ , and  $\Lambda = 0$ . The term under Iverson's brackets contributes, reducing the average depth to

$$\mathbb{E}[D_n] = 2H_{n-1} - 1 - 2 \sum_{i=3}^n \frac{1}{i(i-1)} = 2H_{n-1} - 1 - 2\left(\frac{1}{2} - \frac{1}{n}\right).$$

□

**Corollary 3.2.** *In a uniform recursive tree, for  $n \geq 1$ , we have*

$$\mathbb{E}[D_n] = H_{n-1}.$$

*Proof.* Omitted, as the result is well known (see [9], for example). □

**Corollary 3.3.** *In an old-age recursive tree, for  $n \geq 2$ , we have*

$$\mathbb{E}[D_n] = \frac{2}{3}H_{n-1} + \frac{1}{3}.$$

*Proof.* In an old-age recursive tree, we have  $\alpha > 0$ ,  $\beta = -\alpha$ , and  $\Lambda = 2$ , and the term under Iverson's brackets vanishes, leaving  $\mathbb{E}[D_n] = \frac{2}{3}H_{n-1} + \frac{1}{3}$ , for  $n \geq 3$ . We note that the formula applies at  $n = 2$ , as well.  $\square$

In the sequel, there are numerous occasions to use asymptotics of the gamma function. In particular, we appeal to Stirling approximations. The following sharp bound is also helpful.

**Lemma 3.1.** *Let  $\Lambda$  be in the range  $[0, 2]$ . So,  $\Gamma(i - \Lambda)$  is well defined for  $i \geq 3$ . We have*

$$\frac{\Gamma(i - \Lambda)}{\Gamma(i + 1)} \leq \frac{3^\Lambda}{2i^{\Lambda+1}} \Gamma(3 - \Lambda) =: \frac{K}{i^{\Lambda+1}}.$$

*Proof.* We prove the assertion by induction. At the basis ( $i = 3$ ), we have equality, and the inequality is true. As an induction hypothesis, assume the inequality is true for some  $i \geq 3$ . Using the hypothesis in the induction step, we get

$$\frac{\Gamma(i + 1 - \Lambda)}{\Gamma(i + 2)} \leq \frac{(i - \Lambda) \Gamma(i - \Lambda)}{(i + 1) \Gamma(i + 1)} \leq K \frac{i - \Lambda}{(i + 1)i^{\Lambda+1}}.$$

Note that

$$\frac{i - \Lambda}{(i + 1)i^{\Lambda+1}} \leq \frac{1}{(i + 1)^{\Lambda+1}},$$

as can be seen from the reorganization

$$1 - \frac{\Lambda}{i} \leq \left(\frac{i}{i + 1}\right)^\Lambda \leq 1 - \frac{\Lambda}{i} + \frac{\Lambda(\Lambda + 1)}{2i^2}.$$

So, we have

$$\frac{\Gamma(i + 1 - \Lambda)}{\Gamma(i + 2)} \leq \frac{K}{(i + 1)^{\Lambda+1}},$$

completing the induction.  $\square$

**Proposition 3.2.** *Let  $D_n$  be the depth of node  $n$  in a preferential recursive tree of age index  $\Lambda$ . As  $n \rightarrow \infty$ , we have*

$$\mathbb{E}[D_n] = \frac{2}{1 + \Lambda} H_{n-1} + C_\Lambda + O\left(\frac{1}{n^{\Lambda+1}}\right),$$

where

$$C_\Lambda = \frac{\Lambda - 1}{\Lambda + 1} + [[\Lambda \neq 2]] \frac{2(\Lambda - 1)}{(\Lambda + 1) \Gamma(2 - \Lambda)} \sum_{i=3}^{\infty} \frac{\Gamma(i - \Lambda)}{(i - 1) \Gamma(i + 1)}.$$

*Proof.* Consider the sum in the mean depth in Proposition 3.1. Let us call that sum,  $S_n$ . We prove that  $S_n$  converges to a limit, as  $n \rightarrow \infty$ . With  $i$  starting at 3 in the sum and  $\Lambda \in [0, 2]$ , the argument  $i - \Lambda$  is in the increasing range of the gamma function. So, we have

$$\begin{aligned} S_n &= \sum_{i=3}^n \frac{\Gamma(i - \Lambda)}{(i - 1) \Gamma(i + 1)} \\ &\leq \sum_{i=3}^n \frac{\Gamma(i)}{(i - 1) \Gamma(i + 1)} \\ &= \sum_{i=3}^n \frac{1}{(i - 1)i} \\ &< \sum_{i=2}^n \frac{1}{(i - 1)^2} \\ &< \frac{\pi^2}{6}. \end{aligned}$$

With the summand in  $S_n$  being positive, the series  $S_n$  is increasing (in  $n$ ). Thus,  $S_n$  is an increasing bounded sequence and converges to a limit, say  $S_\Lambda^*$ .

To determine the rate of that convergence, we use the inequality in Lemma 3.1 in the tail of the series

$$S_\Lambda^* - S_n = \sum_{i=n+1}^{\infty} \frac{\Gamma(i - \Lambda)}{(i - 1) \Gamma(i + 1)} < \sum_{i=n+1}^{\infty} \frac{i^{-\Lambda-1}}{i - 1} = O\left(\frac{1}{n^{\Lambda+1}}\right).$$

Inserting the asymptotics in the exact expression, we get

$$\mathbb{E}[D_n] = \frac{2}{\Lambda + 1} H_{n-1} + \frac{\Lambda - 1}{\Lambda + 1} + [[\Lambda \neq 2]] \frac{2(\Lambda - 1)}{(\Lambda + 1) \Gamma(2 - \Lambda)} \left( S_\Lambda^* + O\left(\frac{1}{n^{\Lambda+1}}\right) \right).$$

Take the factor  $\frac{\Lambda-1}{\Lambda+1} + [[\Lambda \neq 2]] \frac{2(\Lambda-1)}{(\Lambda+1)\Gamma(2-\Lambda)}$  to be  $C_\Lambda$ , and the asymptotic form in the proposition follows.  $\square$

### 3.2 The variance of the depth

We start with the second moment of  $D_n$ .

**Proposition 3.3.** *Let  $D_n$  be the depth of node  $n$  in a preferential recursive tree of age index  $\Lambda$ . We have  $\mathbb{V}\text{ar}[D_1] = \mathbb{V}\text{ar}[D_2] = 0$ , and, for  $n \geq 3$ , we have*

$$\begin{aligned} \mathbb{V}\text{ar}[D_n] = & 3 - \frac{\Lambda}{2} + \sum_{j=4}^n \frac{2 [[\Lambda \neq 2]] \Gamma(j - \Lambda)}{\Gamma(2 - \Lambda)(j - 1) \Gamma(j + 1)} + \frac{2 \Gamma(j - \Lambda)}{(j - 1) \Gamma(j + 1)} \sum_{i=3}^j \frac{c_i \Gamma(i)}{\Gamma(i - \Lambda)} \\ & - \left( \frac{2}{\Lambda + 1} H_{n-1} + \frac{\Lambda - 1}{\Lambda + 1} \right. \\ & \left. + [[\Lambda \neq 2]] \frac{2(\Lambda - 1)}{(\Lambda + 1) \Gamma(2 - \Lambda)} \sum_{i=3}^n \frac{\Gamma(i - \Lambda)}{(i - 1) \Gamma(i + 1)} \right)^2, \end{aligned}$$

with

$$\begin{aligned} c_i = & \frac{((2 - \Lambda)i - 2 + 2\Lambda) \Gamma(i - 1 - \Lambda)}{(i - 2) \Gamma(-i\Lambda - 1)} + 2 \sum_{k=3}^{i-2} \frac{\Gamma(k - \Lambda)}{(k - 1) \Gamma(k + 1)} \\ & + \frac{4}{\Lambda + 1} H_i + \frac{(4 - 2\Lambda)i}{(\Lambda + 1)(i - 1)} + \frac{(2\Lambda - 8)i}{(\Lambda + 1)(i - 1)} + \frac{3\Lambda - 1}{\Lambda - 1}. \end{aligned}$$

*Proof.* To a large extent, the derivation follows that for the mean, and we only highlight the major steps. We first take the second derivative of (5) and evaluate it at  $t = 0$ . This yields a recurrence for the second moment of the depth, involving the exact mean of  $D_{n-1}$  and  $D_{n-2}$ , which we now have from Proposition 3.1:

$$\begin{aligned} & \binom{n}{2} \mathbb{E}[D_n^2] - \left( 2 \binom{n-1}{2} + \left(1 - \frac{\Lambda}{2}\right)n - 1 + \Lambda \right) \mathbb{E}[D_{n-1}^2] \\ & + \left( \binom{n-2}{2} + \left(1 - \frac{\Lambda}{2}\right)(n-2) \right) \mathbb{E}[D_{n-1}^2] \\ & - 2 \left( \left(1 - \frac{\Lambda}{2}\right)n - 1 + \Lambda \right) \mathbb{E}[D_{n-1}] + 2 \left(1 - \frac{\Lambda}{2}\right)(n-2) \mathbb{E}[D_{n-2}] \\ & - \left( \left(1 - \frac{\Lambda}{2}\right)n - 1 + \Lambda \right) + \left(1 - \frac{\Lambda}{2}\right)(n-1) = 0. \end{aligned}$$

Again, the identity (8) reorganizes the recurrence in the iterable form

$$\begin{aligned}
& \binom{n}{2} (\mathbb{E}[D_n^2] - \mathbb{E}[D_{n-1}^2]) \\
& - \left( \binom{n-2}{2} + \left(1 - \frac{\Lambda}{2}\right)(n-2) \right) (\mathbb{E}[D_{n-1}^2] - \mathbb{E}[D_{n-2}^2]) \\
& = 2 \left( \left(1 - \frac{\Lambda}{2}\right)n - 1 + \Lambda \right) \mathbb{E}[D_{n-1}] - 2 \left(1 - \frac{\Lambda}{2}\right)(n-2) \mathbb{E}[D_{n-2}] \\
& \quad + \left( \left(1 - \frac{\Lambda}{2}\right)n - 1 + \Lambda \right) - \left(1 - \frac{\Lambda}{2}\right)(n-2) \\
& = 2 \left( \left(1 - \frac{\Lambda}{2}\right)n - 1 + \Lambda \right) \mathbb{E}[D_{n-1}] \\
& \quad - 2 \left(1 - \frac{\Lambda}{2}\right)(n-2) \mathbb{E}[D_{n-2}] + 1,
\end{aligned}$$

for  $n \geq 4$ .

Let us call the difference  $\mathbb{E}[D_n^2] - \mathbb{E}[D_{n-1}^2]$  by the name  $b_n$ , and call the toll function on the right-hand side  $c_n$ . This linearizes the recurrence in the form

$$b_n = \frac{(n-2)(n-1-\Lambda)}{n(n-1)} b_{n-1} + \frac{2c_n}{n(n-1)}.$$

This form is similar to that in (9)—the one difference being that the toll function is more involved. Tracing through the solution of the mean, we construct a similar solution for the second moment:

$$\begin{aligned}
b_n &= \mathbb{E}[D_n^2] - \mathbb{E}[D_{n-1}^2] \\
&= [[\Lambda \neq 2]] \frac{2\Gamma(n-\Lambda)}{\Gamma(2-\Lambda)(n-1)\Gamma(n+1)} + \frac{2\Gamma(n-\Lambda)}{(n-1)\Gamma(n+1)} \sum_{i=3}^n \frac{c_i \Gamma(i)}{\Gamma(i-\Lambda)}.
\end{aligned}$$

Iterating this recurrence back to the boundary condition  $\mathbb{E}[D_3^2]$ , we get

$$\mathbb{E}[D_n^2] = \mathbb{E}[D_3^2] + \sum_{j=4}^n \left( \frac{2 [[\Lambda \neq 2]] \Gamma(j-\Lambda)}{\Gamma(2-\Lambda)(j-1)\Gamma(j+1)} + \frac{2\Gamma(j-\Lambda)}{(j-1)\Gamma(j+1)} \sum_{i=4}^j \frac{c_i \Gamma(i)}{\Gamma(i-\Lambda)} \right).$$

To simplify  $c_n$ , we identify three parts. When we substitute for the expected depth, both  $\mathbb{E}[D_{n-1}]$  and  $\mathbb{E}[D_{n-2}]$  have sums of ratios of gamma functions. However, these sums are telescopic. A second part has harmonic numbers: We can write the harmonic number in  $H_{n-3}$  that appears in  $\mathbb{E}[D_{n-2}]$  as  $H_{n-2} - 1/(n-2)$ . The third part involves the constants.

Putting the parts together, we obtain

$$c_n = \frac{4}{\Lambda + 1} H_{n-2} + \frac{\Lambda + 3}{\Lambda + 1} + \frac{2(\Lambda - 1)((2 - \Lambda)n - 2 + 2\Lambda) \Gamma(n - 1 - \Lambda)}{(\Lambda + 1)(n - 2) \Gamma(2 - \Lambda) \Gamma(n)} \\ + \frac{4(\Lambda - 1)}{(\Lambda + 1) \Gamma(2 - \Lambda)} \sum_{i=3}^{n-2} \frac{\Gamma(i - \Lambda)}{(i - 1) \Gamma(i + 1)}.$$

With  $\mathbb{E}[D_3^2] = 3 - 1/2\Lambda$ , we compute the variance from  $\mathbb{E}[D_n^2] - \mathbb{E}^2[D_n]$ .  $\square$

In studying the mean, we identified the cases  $\Lambda \in \{0, 1, 2\}$  as conditions where we can develop closed forms for the mean; see Corollaries 3.1–3.3. The situation is similar for the variance for these values of  $\Lambda$ , but with more intense derivations. We shall only sketch these derivations. In the ensuing corollaries, we use the following identity in the intermediate steps:

$$\sum_{k=1}^n \frac{H_k}{k} = \frac{H_n^2 + H_n^{(2)}}{2}. \quad (10)$$

**Corollary 3.4.** *In a young-age preferential recursive tree, for  $n \geq 1$ , we have*

$$\mathbb{V}\text{ar}[D_n] = 2H_n - 4H_n^{(2)} + 2.$$

*Proof.* In a young-age preferential tree, we have  $\Lambda = 0$ . We find

$$c_n = 4H_{n-1} + \frac{n-5}{n-1}.$$

Applying Proposition 3.3, after some lengthy cancellations the formula yields

$$\mathbb{E}[D_n^2] = 4H_n^2 - 6H_n - 4H_n^{(2)} + 6.$$

The stated variance follows from  $\mathbb{E}[D_n^2] - \mathbb{E}^2[D_n]$  after some more lengthy computations via (10) and Corollary 3.1.  $\square$

**Corollary 3.5.** *In a uniform recursive tree, for  $n \geq 1$ , we have*

$$\mathbb{V}\text{ar}[D_n] = H_{n-1}^2 - H_{n-1}^{(2)}.$$

*Proof.* Omitted, as the result is well known (see [9], for example).  $\square$

**Corollary 3.6.** *In an old-age recursive tree, for  $n \geq 3$ , the variance is*

$$\mathbb{V}\text{ar}[D_n] = \frac{10}{27} H_{n-1} + \frac{12}{27} H_n^{(2)} - \frac{28}{27} + \frac{12}{27n}.$$

*Proof.* In an old-age recursive tree  $\Lambda$  is 2. The parts with Iverson's brackets vanish in both the mean and the second moment. The second moment in Proposition (3.3) is

$$\mathbb{E}[D_n^2] = \frac{4}{9} H_{n-1}^2 + \frac{22}{27} H_{n-1} + \frac{4}{9} H_{n-1}^{(2)} - \frac{25}{27} + \frac{4}{9n}.$$

The stated variance follows from  $\mathbb{E}[D_n^2] - \mathbb{E}^2[D_n]$  after some more lengthy computations via (10) and the exact mean in Corollary 3.3.  $\square$

In Corollaries 3.4–3.6, the asymptotic order of the variance is  $\ln n$ , although the second moment is of order  $(\ln n)^2$ . This cancellation occurs because the leading term in  $\mathbb{E}[D_n]^2$  matches exactly the leading term in  $\mathbb{E}[D_n^2]$ . The same phenomenon holds for every  $\Lambda \in [0, 2]$ ; the absence of a closed form for non-integer  $\Lambda$  merely makes the bookkeeping more delicate.

Having observed this cancellation for the three classical integer values of  $\Lambda$ , we next extract the asymptotics of  $\mathbb{E}[D_n^2]$  down to the term of order  $\ln n$  beyond  $4H_n^2/(\Lambda+1)^2$ , so as to identify the variance coefficient in the general case.

**Corollary 3.7.** *Let  $D_n$  be the depth of node  $n$  in a preferential recursive tree of age index  $\Lambda$ . As  $n \rightarrow \infty$ , the variance of  $D_n$  is*

$$\mathbb{V}\text{ar}[D_n] = \frac{2 + 8\Lambda - 2\Lambda^2}{(\Lambda + 1)^3} H_{n-1} + O(1).$$

*Proof.* The case  $\Lambda = 2$  is handled in Corollary 3.6. So, we assume  $0 \leq \Lambda < 2$ .

We first develop an asymptotic equivalent to  $c_i$ . The sum in  $c_i$  is  $S_i$  that appears in the proof of Proposition 3.1. We saw there that that sum approaches a limit  $S_\Lambda^*$  at a rate that is  $O(1/i^{\Lambda+1})$ . So, for large  $i$ , we have

$$\begin{aligned} c_i &= \frac{4}{\Lambda + 1} \left( H_i - \frac{1}{i} + \frac{1}{i^2} \right) + \frac{4(\Lambda - 1)}{(\Lambda + 1) \Gamma(\Lambda - 2)} \left( S^* + O\left(\frac{1}{i^{\Lambda+1}}\right) \right) \\ &\quad + \frac{\Lambda + 3}{\Lambda + 1} + O\left(\frac{1}{i^{\Lambda+2}}\right) \\ &= \frac{4}{\Lambda + 1} H_i + \frac{4(\Lambda - 1)}{(\Lambda + 1) \Gamma(2 - \Lambda)} S_\Lambda^* + \frac{\Lambda + 3}{\Lambda + 1} + O\left(\frac{1}{i}\right). \end{aligned}$$

Let us call the constant term in this expression  $q_\Lambda$ .

Plugging this asymptotic estimate in the inner sum of the variance, while replacing the ratio of the gamma functions with their well-known expansion, we get for the inner sum

$$\begin{aligned} \sum_{i=3}^j \left( i^\Lambda - \frac{\Lambda(\Lambda+1)}{2} i^{\Lambda-1} + O(i^{\Lambda-2}) \right) \left( \frac{4}{\Lambda+1} \ln i + q_\Lambda + \frac{4\gamma}{\Lambda+1} + O\left(\frac{1}{i}\right) \right) \\ = \sum_{i=3}^n \left( \frac{4}{\Lambda+1} i^\Lambda \ln i + \left( q_\Lambda + \frac{4\gamma}{\Lambda+1} \right) i^\Lambda + O(i^{\Lambda-1}) \right). \end{aligned}$$

Euler-Maclaurin approximation by integrals is amenable to the remaining sum, giving for the inner sum

$$\begin{aligned} \left( \frac{4}{(\Lambda+1)^2} j^{\Lambda+1} \ln j - \frac{4}{(\Lambda+1)^3} j^{\Lambda+1} + O(j^\Lambda) \right) \\ + \left( q_\Lambda + \frac{4\gamma}{\Lambda+1} \right) \left( \frac{j^{\Lambda+1}}{\Lambda+1} + O(j^\Lambda) \right) + O(j^\Lambda) \\ = \frac{4}{(\Lambda+1)^2} j^{\Lambda+1} \ln j + \left( \left( \frac{q_\Lambda}{\Lambda+1} + \frac{4\gamma}{(\Lambda+1)^2} \right) - \frac{4}{(\Lambda+1)^3} \right) j^{\Lambda+1} + O(j^\Lambda). \end{aligned}$$

Let us call the term  $\frac{q_\Lambda}{\Lambda+1} + \frac{4\gamma}{(\Lambda+1)^2} - \frac{4}{(\Lambda+1)^3}$  as  $r_\Lambda$ . We are now ready to execute the outer summation. Toward an approximation by an integral, we write the summand with the ratio of gamma functions expressed in asymptotic form:

$$\begin{aligned} \mathbb{E}[D_n^2] &= 2 \sum_{j=4}^n \left( \frac{4}{(\Lambda+1)^2} j^{\Lambda+1} \ln j + r_\Lambda j^{\Lambda+1} + O(j^\Lambda) \right) \\ &\quad \times \left( j^{-\Lambda-2} + \frac{\Lambda(\Lambda+1)}{2} j^{\Lambda-3} + O(j^{-\Lambda-4}) \right) \\ &= \frac{8}{(\Lambda+1)^2} \sum_{j=4}^n \frac{\ln j}{j} + 2r_\Lambda \sum_{j=4}^n \frac{1}{j} + O\left(\frac{\ln j}{j^2}\right) \\ &= \frac{8}{(\Lambda+1)^2} \left( \left( \int_{j=4}^n \frac{\ln x}{x} dx \right) + O\left(\frac{\ln n}{n}\right) \right) + 2r_\Lambda H_n + O\left(\frac{\ln n}{n}\right) \\ &= \frac{4}{(\Lambda+1)^2} \ln^2 n + 2r_\Lambda \left( H_n + O\left(\frac{1}{n}\right) \right) + O\left(\frac{\ln n}{n}\right) \\ &= \frac{4}{(\Lambda+1)^2} \left( H_{n-1} - \gamma + O\left(\frac{1}{n}\right) \right)^2 + 2r_\Lambda H_{n-1} + O\left(\frac{\ln n}{n}\right) \end{aligned}$$

$$= \frac{4}{(\Lambda + 1)^2} H_{n-1}^2 + \left( 2r_\Lambda - \frac{8\gamma}{(\Lambda + 1)^2} \right) H_{n-1} + O\left(\frac{\ln n}{n}\right).$$

Squaring the asymptotic form in Proposition 3.2, we get

$$\mathbb{E}^2[D_n] = \frac{4}{(1 + \Lambda)^2} H_{n-1}^2 + \frac{4C_\Lambda}{\Lambda + 1} H_{n-1} + O(1).$$

After cancellation of the leading terms in  $\mathbb{E}[D_n^2]$  and in  $\mathbb{E}[D_n]^2$ , we find the variance

$$\begin{aligned} \mathbb{V}\text{ar}[D_n] &= \left( 2r_\Lambda - \frac{8\gamma}{(\Lambda + 1)^2} - \frac{4C_\Lambda}{\Lambda + 1} \right) H_{n-1} + O(1) \\ &= \left( 2\left( \frac{q_\Lambda}{\Lambda + 1} + \frac{4\gamma}{(\Lambda + 1)^2} - \frac{4}{(\Lambda + 1)^3} \right) - \frac{8\gamma}{(\Lambda + 1)^2} - \frac{4C_\Lambda}{\Lambda + 1} \right) H_{n-1} + O(1) \\ &= \left( 2\left( \frac{4(\Lambda - 1)}{(\Lambda + 1)\Gamma(\Lambda - 2)} \right) S_\Lambda^* + \frac{\Lambda + 3}{(\Lambda + 1)^2} \right) - \frac{8}{(\Lambda + 1)^3} - \frac{4C_\Lambda}{\Lambda + 1} \Big) H_{n-1} \\ &\quad + O(1). \end{aligned}$$

Recalling the relation between  $C_\Lambda$  and  $S_\Lambda^*$ , a further cancellation of the terms involving  $S_\Lambda^*$  yields

$$\begin{aligned} \mathbb{V}\text{ar}[D_n] &= \left( \frac{2(\Lambda + 3)}{(\Lambda + 1)^2} - \frac{8}{(\Lambda + 1)^3} - \frac{4(\Lambda - 1)}{(\Lambda + 1)^2} \right) H_{n-1} + O(1) \\ &= \frac{2 + 8\Lambda - 2\Lambda^2}{(\Lambda + 1)^3} H_{n-1} + O(1). \end{aligned}$$

□

Since  $\mathbb{V}\text{ar}[D_n] = O(\ln n)$  while  $\mathbb{E}[D_n] \sim a \ln n$  with  $a = 2/(1 + \Lambda)$ , Chebyshev's inequality yields the following weak law.

**Corollary 3.8.** *As  $n \rightarrow \infty$ , we have*

$$\frac{D_n}{\ln n} \xrightarrow{\mathbb{P}} \frac{2}{\Lambda + 1}.$$

### 3.3 Asymptotic distribution

The preceding subsections determine the growth of the first two moments of  $D_n$ : both the mean and the variance are of exact order  $\ln n$  (Proposition 3.2

and Corollary 3.7). We now turn to the limiting distribution. Set

$$\mu_n := \mathbb{E}[D_n], \quad \Delta_n := D_n - \mu_n, \quad a := \frac{2}{1 + \Lambda}, \quad V_\Lambda := \frac{2 + 8\Lambda - 2\Lambda^2}{(\Lambda + 1)^3},$$

so that

$$\mathbb{V}\text{ar}[D_n] = V_\Lambda H_{n-1} + O(1) \sim V_\Lambda \ln n. \quad (11)$$

**Theorem 3.1.** *Let  $\Lambda \in [0, 2]$ . As  $n \rightarrow \infty$ ,*

$$\frac{D_n - \mu_n}{\sqrt{\ln n}} \xrightarrow{\mathcal{L}} \mathcal{N}(0, V_\Lambda).$$

*Equivalently,*

$$\frac{D_n - \mu_n}{\sqrt{\mathbb{V}\text{ar}[D_n]}} \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1).$$

The argument is organised as follows. We begin with the parent-index decomposition  $\Delta_n = \Delta_n^{(1)} + \Delta_n^{(2)}$ , identify the limiting law of the normalised parent, and show that the mean-fluctuation term  $\Delta_n^{(2)}$  is negligible on the scale  $\sqrt{\ln n}$ . An averaging lemma for the attachment kernel is then recorded. With these tools in hand, we prove by induction that every moment of  $\Delta_n/\sqrt{\ln n}$  converges to the corresponding moment of  $\mathcal{N}(0, V_\Lambda)$ . The central limit theorem follows from the Fréchet–Shohat theorem. Throughout, the decomposition plays an essential role: it yields the mixture recurrence for the moments, and the identity  $\mathbb{E}[W^2] = -V_\Lambda \mathbb{E}[\ln J]$  derived from the variance asymptotics converts that recurrence into the Wick relations of the Gaussian law.

### Decomposition and the parent index

Since  $D_n = 1 + D_{J_n}$  and  $\mu_n = 1 + \sum_{k=1}^{n-1} \mu_k p_{n,k}$ , with

$$p_{n,k} := \mathbb{P}(\mathcal{A}_{n,k}) = \frac{\Lambda n + 2(1 - \Lambda)k}{n(n - 1)},$$

and since the affinities (1) depend only on the labels, the parent  $J_n$  is independent of  $(D_1, \dots, D_{n-1})$ . Centring therefore gives the exact decomposition

$$\Delta_n = \Delta_{J_n} + (\mu_{J_n} - \mathbb{E}[\mu_{J_n}]) = \Delta_n^{(1)} + \Delta_n^{(2)}, \quad (12)$$

where

$$\Delta_n^{(1)} := \Delta_{J_n} = \sum_{k=1}^{n-1} \Delta_k \mathbb{I}_{\mathcal{A}_{n,k}}, \quad (13)$$

$$\Delta_n^{(2)} := \mu_{J_n} - \mathbb{E}[\mu_{J_n}] = \sum_{k=1}^{n-1} \mu_k (\mathbb{I}_{\mathcal{A}_{n,k}} - p_{n,k}). \quad (14)$$

Writing  $\nu_{n,k} := \mu_k - \mathbb{E}[\mu_{J_n}]$  (so that  $\sum_k p_{n,k} \nu_{n,k} = 0$  and  $\Delta_n^{(2)} = \nu_{n,J_n}$ ), the binomial theorem yields, for every integer  $m \geq 1$ ,

$$M_n(m) := \mathbb{E}[\Delta_n^m] = \sum_{k=1}^{n-1} p_{n,k} M_k(m) + R_n(m), \quad (15)$$

with

$$R_n(m) := \sum_{j=0}^{m-1} \binom{m}{j} \sum_{k=1}^{n-1} p_{n,k} M_k(j) \nu_{n,k}^{m-j}. \quad (16)$$

When  $m = 2$ , this is the familiar variance decomposition  $\mathbb{V}\text{ar}[D_n] = \sum_k p_{n,k} \mathbb{V}\text{ar}[D_k] + \mathbb{E}[(\Delta_n^{(2)})^2]$ .

**Lemma 3.2.** *As  $n \rightarrow \infty$ , one has  $J_n/n \xrightarrow{\mathcal{L}} J$ , where  $J$  is supported on  $(0, 1)$  and admits the density*

$$f_\Lambda(x) = \Lambda + 2(1 - \Lambda)x, \quad x \in (0, 1). \quad (17)$$

*In particular,  $\mathbb{P}(J > 0) = 1$  and  $\mathbb{E}[|\ln J|^q] < \infty$  for every  $q \geq 1$ .*

*Proof.* For fixed  $x \in (0, 1)$  and  $m := \lfloor xn \rfloor$ ,

$$\mathbb{P}\left(\frac{J_n}{n} \leq x\right) = \sum_{k=1}^m \frac{\Lambda n + 2(1 - \Lambda)k}{n(n-1)} = \frac{\Lambda m}{n-1} + \frac{(1 - \Lambda)m(m+1)}{n(n-1)}.$$

Substituting  $m = xn + O(1)$  gives  $\Lambda x + (1 - \Lambda)x^2 + o(1)$ , which is the distribution function of the claimed law. Near the origin one has  $f_\Lambda(x) \asymp \Lambda + 2x$ , and  $|\ln x|^q$  is integrable on  $(0, 1)$  against any such density; hence all logarithmic moments of  $J$  are finite.  $\square$

**Lemma 3.3.** *As  $n \rightarrow \infty$ ,*

$$\Delta_n^{(2)} \xrightarrow{\mathcal{L}} W := a \ln J - a \mathbb{E}[\ln J]. \quad (18)$$

*Consequently,*

$$\frac{\Delta_n^{(2)}}{\sqrt{\ln n}} \xrightarrow{\mathbb{P}} 0, \quad (19)$$

*and for every integer  $q \geq 1$ ,*

$$\sup_{n \geq 2} \mathbb{E}[|\Delta_n^{(2)}|^q] < \infty. \quad (20)$$

*Moreover, writing  $W_n := \Delta_n^{(2)}$ ,*

$$\mathbb{E}[W^2] = -V_\Lambda \mathbb{E}[\ln J]. \quad (21)$$

*Proof.* Proposition 3.2 yields  $\mu_k = aH_{k-1} + C_\Lambda + O(k^{-(\Lambda+1)})$ , and therefore

$$\mu_k = a \ln k + C_\Lambda + a\gamma + o(1) \quad (k \rightarrow \infty).$$

Hence

$$\Delta_n^{(2)} = a \ln\left(\frac{J_n}{n}\right) - a \mathbb{E}\left[\ln\left(\frac{J_n}{n}\right)\right] + r_n(J_n) - \mathbb{E}[r_n(J_n)],$$

where the remainder  $r_n(k) := \mu_k - a \ln(k/n) - a \ln n$  tends to  $C_\Lambda + a\gamma$  along  $k/n \rightarrow y \in (0, 1)$  and is uniformly integrable under the laws of  $J_n/n$  (using  $\mu_k = O(\ln k)$  together with the polynomial error term). Lemma 3.2 and continuous mapping give (18).

Uniform integrability of  $\{|\ln(J_n/n)|^q\}_{n \geq 2}$  follows from the convergence of the densities of  $J_n/n$  to  $f_\Lambda$  and the integrability of  $|\ln x|^q f_\Lambda(x)$ . This yields (20), from which (19) is immediate.

To obtain (21), combine the variance decomposition with Corollary 3.7. On the one hand,

$$\text{Var}[D_n] = V_\Lambda H_{n-1} + C + o(1) = V_\Lambda \ln n + V_\Lambda \gamma + C + o(1)$$

for a constant  $C$  depending only on  $\Lambda$ . On the other hand,

$$\sum_{k=1}^{n-1} p_{n,k} \text{Var}[D_k] = \sum_{k=1}^{n-1} p_{n,k} (V_\Lambda H_{k-1} + C + o(1)) = V_\Lambda \mathbb{E}[H_{J_n-1}] + C + o(1)$$

$$= V_\Lambda \ln n + V_\Lambda \mathbb{E} \left[ \ln \left( \frac{J_n}{n} \right) \right] + V_\Lambda \gamma + C + o(1).$$

Since  $\mathbb{V}\text{ar}[D_n] = \sum_k p_{n,k} \mathbb{V}\text{ar}[D_k] + \mathbb{E}[W_n^2]$ , subtraction yields

$$\mathbb{E}[W_n^2] = -V_\Lambda \mathbb{E} \left[ \ln \left( \frac{J_n}{n} \right) \right] + o(1).$$

Passing to the limit with the help of (20) gives (21).  $\square$

As an immediate consequence of (12) and (19), the sequences  $\Delta_n/\sqrt{\ln n}$  and  $\Delta_n^{(1)}/\sqrt{\ln n}$  are asymptotically equivalent in probability. It is therefore enough to establish the central limit theorem for  $\Delta_n$ ; the corresponding statement for  $\Delta_n^{(1)}$  follows at once.

### An averaging lemma

The recurrence (15) is driven by the same averaging operator that governs the mean. The following lemma records its mapping properties on slowly varying forcings.

**Lemma 3.4.** *Let  $(u_n)_{n \geq 1}$  satisfy  $u_1 = 0$  and*

$$u_n = \sum_{k=1}^{n-1} p_{n,k} u_k + f_n, \quad n \geq 2. \quad (22)$$

(i) *If  $|f_n| \leq F(1 + \ln n)^\beta$  for constants  $F < \infty$  and  $\beta \geq 0$ , then  $|u_n| \leq CF(1 + \ln n)^{\beta+1}$  for some  $C = C(\Lambda, \beta) < \infty$ .*

(ii) *If  $f_n = o((1 + \ln n)^\beta)$ , then  $u_n = o((1 + \ln n)^{\beta+1})$ .*

*Proof.* Writing (22) at the three consecutive indices  $n$ ,  $n-1$  and  $n-2$ , and eliminating the averaged sums by means of the identity (8), one obtains the first-order recurrence

$$\alpha_n = g_n \alpha_{n-1} + \tilde{h}_n, \quad \alpha_n := u_n - u_{n-1}, \quad g_n = \frac{(n-2)(n-1-\Lambda)}{n(n-1)},$$

in which

$$|\tilde{h}_n| \leq \frac{C'_\Lambda}{n^2} \max\{|f_n|, |f_{n-1}|, |f_{n-2}|\}. \quad (23)$$

(This is the exact analogue of the inhomogeneous term  $2/(n(n-1))$  in (9).) Unrolling from the initial index and using the Gamma-ratio evaluation already exploited in Proposition 3.1,

$$\prod_{j=i+1}^n g_j = \frac{(i-1)i}{(n-1)n} \cdot \frac{\Gamma(n-\Lambda)/\Gamma(i-\Lambda)}{\Gamma(n)/\Gamma(i)} = \Theta\left(\left(\frac{i}{n}\right)^{2+\Lambda}\right), \quad (24)$$

uniformly for  $2 \leq i \leq n$  (cf. Lemma 3.1), we find

$$|\alpha_n| \leq C \sum_{i=2}^n \frac{\max_{i-2 \leq \ell \leq i} |f_\ell|}{i^2} \left(\frac{i}{n}\right)^{2+\Lambda}.$$

For (i), the bound on  $f$  yields  $|\alpha_n| \leq CF(1 + \ln n)^\beta/n$ , and summing the increments gives the asserted estimate on  $u_n$ . In particular, every solution of (22) with  $|f_n| \leq F(1 + \ln n)^\beta$  satisfies

$$|u_n - u_{n-1}| = O\left(\frac{(1 + \ln n)^\beta}{n}\right). \quad (25)$$

In the special case  $f_n \equiv 1$ , one recovers  $u_n = O(\ln n)$ , in agreement with Proposition 3.2.

For (ii), fix  $\varepsilon > 0$  and split  $f_n = f_n^\sharp + f_n^\flat$ , where  $|f_n^\sharp| \leq \varepsilon(1 + \ln n)^\beta$  for all large  $n$  and  $f_n^\flat$  has finite support. The corresponding solutions satisfy  $|u_n^\sharp| \leq C\varepsilon(1 + \ln n)^{\beta+1}$  for large  $n$  by (i), while  $u_n^\flat = O(1)$  because only finitely many increments are nonzero. Hence  $\limsup_n |u_n|/(1 + \ln n)^{\beta+1} \leq C\varepsilon$ . Letting  $\varepsilon \rightarrow 0$  completes the proof.  $\square$

### Convergence of moments

Let  $Z \sim \mathcal{N}(0, V_\Lambda)$  and write  $\mu_m := \mathbb{E}[Z^m]$ , so that  $\mu_m = 0$  for  $m$  odd and  $\mu_m = (m-1)V_\Lambda\mu_{m-2}$  for  $m$  even (Wick's formula), with  $\mu_0 = 1$  and  $\mu_2 = V_\Lambda$ .

**Lemma 3.5.** *For every integer  $m \geq 1$ ,*

$$M_n(m) = \mu_m (\ln n)^{m/2} + o((\ln n)^{m/2}). \quad (26)$$

*If moreover  $m$  is odd, then*

$$M_n(m) = O((\ln n)^{(m-1)/2}). \quad (27)$$

*In particular,  $\mathbb{E}[(\Delta_n/\sqrt{\ln n})^m] \rightarrow \mu_m$ .*

*Proof.* The proof proceeds by induction on  $m$ . Throughout, approximations under the discrete measure  $p_{n,k}$  are justified by splitting at  $k = n^\alpha$  ( $\alpha \in (0, 1)$  fixed) and by the uniform integrability coming from (20) and Lemma 3.2; we never factor a  $k$ -dependent  $O(1)$  out of a signed sum and then invoke  $\mathbb{E}[W_n] = 0$ .

*Base cases.* For  $m = 1$  one has  $M_n(1) = 0 = \mu_1$ . For  $m = 2$ , Corollary 3.7 gives  $M_n(2) = V_\Lambda \ln n + O(1)$ .

*Induction hypothesis.* Fix  $m \geq 3$ , and assume that (26) holds for all orders  $j < m$ , and that (27) holds for all odd orders  $j < m$ . In addition, for every odd order  $\ell < m$  the sequence  $(M_n(\ell))_{n \geq 2}$  solves an averaging recurrence (22) whose forcing is  $O((\ln n)^{(\ell-3)/2})$ ; consequently (25) yields

$$|M_n(\ell) - M_{n-1}(\ell)| = O\left(\frac{(1 + \ln n)^{(\ell-3)/2}}{n}\right). \quad (28)$$

*Even indices in  $R_n(m)$ .* Fix an even index  $j < m$ . Write  $M_k(j) = \mu_j(\ln k)^{j/2} + \varepsilon_k(\ln k)^{j/2}$  with  $\varepsilon_k \rightarrow 0$ . Then

$$\sum_{k=1}^{n-1} p_{n,k} M_k(j) \nu_{n,k}^{m-j} = \mu_j \sum_{k=1}^{n-1} p_{n,k} (\ln k)^{j/2} \nu_{n,k}^{m-j} + \sum_{k=1}^{n-1} p_{n,k} \varepsilon_k (\ln k)^{j/2} \nu_{n,k}^{m-j}. \quad (29)$$

For the error sum, split at  $k = n^\alpha$ . On  $\{k > n^\alpha\}$  one has  $\sup_{k > n^\alpha} |\varepsilon_k| \rightarrow 0$ , while  $\mathbb{E}[|\ln J_n|^{j/2} |W_n|^{m-j}]$  remains bounded by (20) and the integrability of  $|\ln x|^{j/2} f_\Lambda(x)$ . On  $\{k \leq n^\alpha\}$  one has  $|\nu_{n,k}| = O(\ln n)$  and  $M_k(j) = O((\ln n)^{j/2})$ , whence the contribution is  $O((\ln n)^{j/2+m-j}) \mathbb{P}(J_n \leq n^\alpha)$ . Since  $\mathbb{P}(J_n \leq n^\alpha) = O(n^{\alpha-1})$ , this is  $o((\ln n)^{j/2})$ . Thus the second sum on the right-hand side of (29) is  $o((\ln n)^{j/2})$ .

For the main sum, expand

$$(\ln k)^{j/2} = (\ln n)^{j/2} + \frac{j}{2} (\ln n)^{j/2-1} \ln\left(\frac{k}{n}\right) + O\left((\ln n)^{j/2-2} \ln^2\left(\frac{k}{n}\right)\right),$$

valid for  $k \geq 2$ , and average against  $\nu_{n,k}^{m-j}$ . The resulting expectations involve only polynomials in  $\ln(J_n/n)$  and  $W_n$ , which are uniformly integrable. In particular:

- no contribution of order  $(\ln n)^{m/2}$  arises in  $R_n(m)$ , because the factor  $(\ln n)^{j/2} \mathbb{E}[W_n^{m-j}]$  has total logarithmic degree  $j/2 < (m/2)$  whenever  $j \leq m-1$ , while for  $m-j=1$  one has in addition  $\mathbb{E}[W_n] = 0$ ;

- if  $m - j \geq 3$ , the whole sum is  $O((\ln n)^{j/2}) = o((\ln n)^{m/2-1})$ ;
- if  $m - j = 2$  (i.e.  $j = m - 2$ ), one obtains

$$\sum_k p_{n,k} (\ln k)^{(m-2)/2} \nu_{n,k}^2 = (\ln n)^{(m-2)/2} \mathbb{E}[W_n^2] + o((\ln n)^{m/2-1}),$$

and  $\mathbb{E}[W_n^2] \rightarrow \mathbb{E}[W^2]$  by (20).

*Odd indices in  $R_n(m)$ .* If  $j < m$  is odd and  $j \leq m - 3$ , the bound (27) gives  $M_k(j) = O((\ln k)^{(j-1)/2})$ , hence

$$\left| \sum_k p_{n,k} M_k(j) \nu_{n,k}^{m-j} \right| \leq C \mathbb{E}[(\ln J_n)^{(j-1)/2} |W_n|^{m-j}] = O((\ln n)^{(j-1)/2}) = o((\ln n)^{m/2-1}),$$

with no cancellation required.

The remaining odd index is  $j = m - 1$  (so  $m$  is even and  $m - 1$  is odd). Here a crude bound would only yield  $O((\ln n)^{m/2-1})$ , which is the critical order; one cannot replace  $M_k(m - 1)$  by  $O((\ln k)^{(m-2)/2})$  inside the signed sum and then use  $\mathbb{E}[W_n] = 0$ . Instead, since  $\sum_k p_{n,k} \nu_{n,k} = 0$ ,

$$\sum_k p_{n,k} M_k(m-1) \nu_{n,k} = \sum_k p_{n,k} (M_k(m-1) - M_n(m-1)) \nu_{n,k} = \mathbb{E}[(M_{J_n}(m-1) - M_n(m-1)) W_n]. \quad (30)$$

The increment bound (28) at  $\ell = m - 1$  yields, for  $2 \leq k \leq n$ ,

$$\begin{aligned} |M_k(m-1) - M_n(m-1)| &\leq \sum_{i=\min(k,n)+1}^{\max(k,n)} O\left(\frac{(1 + \ln i)^{(m-4)/2}}{i}\right) \\ &\leq C \left| (\ln n)^{(m-2)/2} - (\ln k)^{(m-2)/2} \right| \leq C' (\ln n)^{(m-4)/2} \left| \ln\left(\frac{k}{n}\right) \right|, \end{aligned}$$

the last step by the mean value theorem. Therefore

$$\left| \sum_k p_{n,k} M_k(m-1) \nu_{n,k} \right| \leq C' (\ln n)^{(m-4)/2} \mathbb{E} \left[ \left| \ln\left(\frac{J_n}{n}\right) \right| |W_n| \right] = O((\ln n)^{(m-4)/2}),$$

which is  $o((\ln n)^{m/2-1})$ .

If instead  $m$  is odd (so that  $j = m - 1$  is even), the even-index analysis above applies: the factor  $(\ln n)^{(m-1)/2} \mathbb{E}[W_n]$  vanishes, and the sum is  $O((\ln n)^{(m-3)/2})$ .

*Leading coefficient.* Collecting the contributions, the only term of exact order  $(\ln n)^{m/2-1}$  in  $R_n(m)$  comes from  $j = m - 2$  when  $m - 2$  is even (otherwise  $\mu_{m-2} = 0$ ):

$$\begin{aligned} \binom{m}{2} \sum_k p_{n,k} M_k(m-2) \nu_{n,k}^2 &= \binom{m}{2} \mu_{m-2} (\ln n)^{(m-2)/2} \mathbb{E}[W_n^2] + o((\ln n)^{m/2-1}) \\ &= \binom{m}{2} \mu_{m-2} \mathbb{E}[W^2] (\ln n)^{m/2-1} + o((\ln n)^{m/2-1}). \end{aligned}$$

Thus

$$R_n(m) = \rho_m (\ln n)^{m/2-1} + o((\ln n)^{m/2-1}), \quad \rho_m := \binom{m}{2} \mu_{m-2} \mathbb{E}[W^2]. \quad (31)$$

When  $m$  is odd one has  $\mu_{m-2} = 0$ , so  $\rho_m = 0$  and  $R_n(m) = O((\ln n)^{(m-3)/2})$ .

*The centred recurrence.* Set  $u_n := M_n(m) - \mu_m (\ln n)^{m/2}$ , with the constant  $\mu_m$  to be determined. From (15),

$$u_n = \sum_{k=1}^{n-1} p_{n,k} u_k + R_n(m) + \mu_m \sum_{k=1}^{n-1} p_{n,k} \left( (\ln k)^{m/2} - (\ln n)^{m/2} \right).$$

The last sum is purely deterministic. The expansion of  $(\ln k)^{m/2} - (\ln n)^{m/2}$  yields

$$\sum_k p_{n,k} \left( (\ln k)^{m/2} - (\ln n)^{m/2} \right) = \frac{m}{2} (\ln n)^{m/2-1} \mathbb{E} \left[ \ln \left( \frac{J_n}{n} \right) \right] + O((\ln n)^{m/2-2}),$$

and therefore

$$u_n = \sum_{k=1}^{n-1} p_{n,k} u_k + \left( \rho_m + \mu_m \cdot \frac{m}{2} \mathbb{E} \left[ \ln \left( \frac{J_n}{n} \right) \right] + o(1) \right) (\ln n)^{m/2-1}. \quad (32)$$

*Choice of  $\mu_m$  and conclusion of the induction.* Since  $\mathbb{E}[\ln J] < 0$ , the equation

$$\rho_m + \mu_m \cdot \frac{m}{2} \mathbb{E}[\ln J] = 0 \quad (33)$$

admits the unique solution

$$\mu_m = - \frac{(m-1) \mu_{m-2} \mathbb{E}[W^2]}{\mathbb{E}[\ln J]}. \quad (34)$$

With this choice, the bracket in (32) tends to zero.

If  $m$  is odd, then  $\rho_m = 0$ , so  $\mu_m = 0$  and the forcing is  $O((\ln n)^{(m-3)/2})$ . Lemma 3.4(i) yields  $M_n(m) = O((\ln n)^{(m-1)/2})$ , which is (27) and implies (26). The same application of Lemma 3.4 gives the increment bound (28) at  $\ell = m$ , closing the induction for odd orders.

If  $m$  is even, the forcing is  $o((\ln n)^{m/2-1})$ , and Lemma 3.4(ii) gives  $u_n = o((\ln n)^{m/2})$ . It remains only to check that (34) agrees with  $\mathbb{E}[Z^m]$ . Substituting the identity (21) into (34) produces

$$\mu_m = (m-1) V_\Lambda \mu_{m-2}.$$

By the induction hypothesis,  $\mu_{m-2} = \mathbb{E}[Z^{m-2}]$ , and the displayed relation is Wick's recurrence for  $\mathcal{N}(0, V_\Lambda)$ . Hence  $\mu_m = \mathbb{E}[Z^m]$ , and the induction is complete.  $\square$

## Conclusion

*Proof of Theorem 3.1.* Lemma 3.5 asserts that

$$\mathbb{E} \left[ \left( \frac{\Delta_n}{\sqrt{\ln n}} \right)^m \right] \rightarrow \mathbb{E}[Z^m] \quad \text{for every integer } m \geq 1,$$

where  $Z \sim \mathcal{N}(0, V_\Lambda)$ . The Gaussian law is moment-determined (Carleman's criterion). By the Fréchet–Shohat theorem,

$$\frac{\Delta_n}{\sqrt{\ln n}} \xrightarrow{\mathcal{L}} \mathcal{N}(0, V_\Lambda).$$

Since  $\text{Var}[D_n]/\ln n \rightarrow V_\Lambda$ , Slutsky's lemma yields the standardised form as well.

Finally, the equivalence in probability between  $\Delta_n/\sqrt{\ln n}$  and  $\Delta_n^{(1)}/\sqrt{\ln n}$  noted after Lemma 3.3 shows that

$$\frac{\Delta_n^{(1)}}{\sqrt{\ln n}} \xrightarrow{\mathcal{L}} \mathcal{N}(0, V_\Lambda).$$

$\square$

**Remark 3.3.** For the classical values  $\Lambda \in \{0, 1, 2\}$  one recovers  $V_0 = 2$ ,  $V_1 = 1$  and  $V_2 = 10/27$ , in agreement with Corollaries 3.4–3.6. In particular, in the uniform case  $\Lambda = 1$ ,  $(D_n - H_{n-1})/\sqrt{\ln n} \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1)$ .

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